1777109511

# SDM Institute of Technology, Ujire
# Department of Information Science and Engineering
# Major-Project Synopsis
# 6th Semester ISE-2026

| | |
|---|---|
| **Title** | Design and Implementation of a Cloud-First Thin-Client Smart Keyboard Workstation |
| **Abstract** | This project proposes a cloud-first smart keyboard workstation for portable web development and automation. A Raspberry Pi operates as a thin client with a built-in mini projector for compact visual output. All development and computational tasks are handled through a cloud-hosted environment with AI-assisted automation. The system provides a low-cost, low-power, and scalable solution for student and field-based use. |
| **Objectives** | • To design a compact smart keyboard workstation using a Raspberry Pi and mini projector.<br>• To enable cloud-based development through a thin-client interface.<br>• To integrate AI-assisted agents for automated coding and task management.<br>• To develop a low-cost, low-power, and scalable portable development system. |
| **Methodology** | The system is built using a Raspberry Pi configured with a lightweight Linux operating system and connected to a compact mini projector for display. A browser-based thin-client interface is used to access a cloud-hosted development environment over the internet. AI-assisted automation agents are deployed in the cloud to perform coding, testing, and task execution. The entire setup is optimized to minimize power consumption, hardware cost, and local processing load. |
| **Proposed outcome** | The proposed system delivers a functional portable workstation for web development using minimal local hardware. It enables users to access powerful cloud-based development tools through a compact smart keyboard interface. AI-assisted automation improves productivity by handling repetitive coding and task management operations. The design significantly reduces power consumption and hardware cost compared to traditional laptops. The solution demonstrates a scalable and practical model for cloud-first development environments. |
| **Team members with phone number** | Harshith(4SU23IS016) – 7019146052<br>Priya Bhargavi (4SU23IS033) - 9380844898<br>Rakshitha C R (4SU23IS036) - 7483005435<br>Roopesh H N (4SU23IS038) - 7676324959 |

**Ms. Leema Rachel Mathew**
(Major project Co-ordinator)

**Dr. G P Hegde**
(HoD)